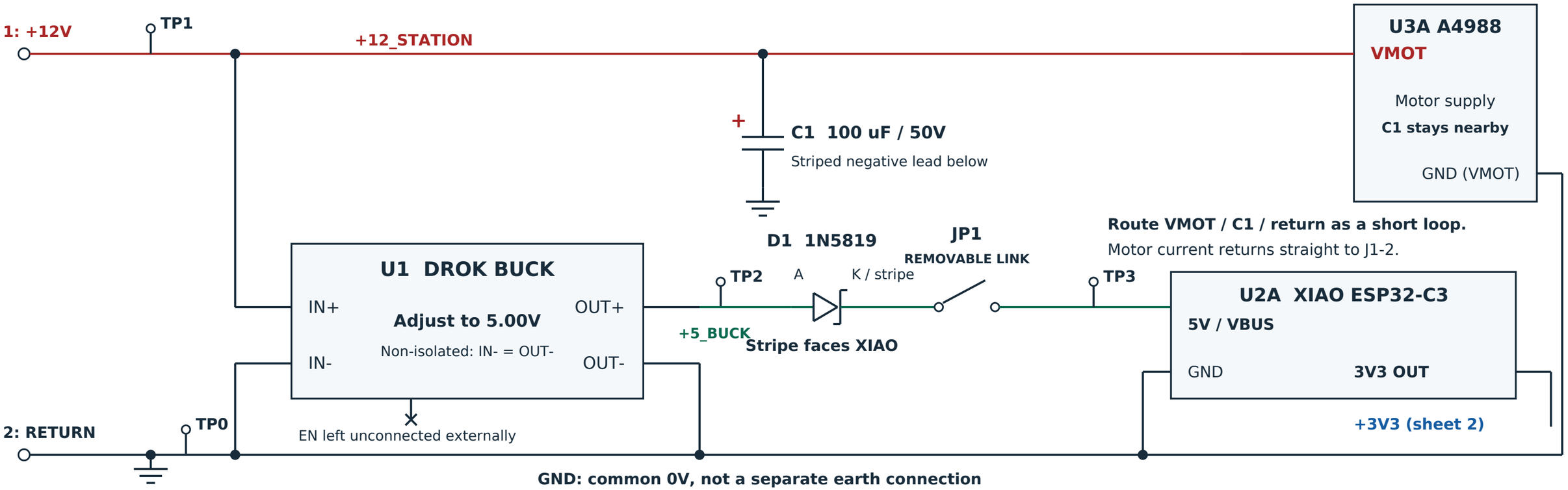
Power circuit - one station

Real wire connections, component symbols, and named nets. Filled dot = connected junction. Same net label = same electrical node.

J1 TWO-WIRE INPUT

Branch fuse is at supply end (see below).



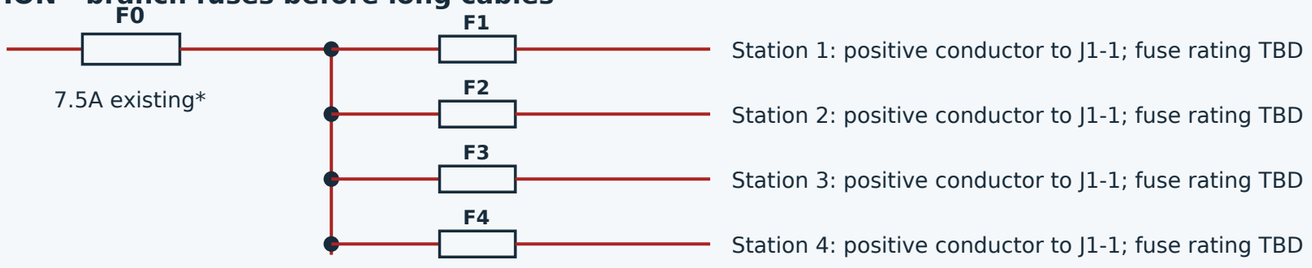
USB / EXTERNAL POWER RULE

USB plugged in: JP1 OPEN. Standalone from 12V: USB unplugged, JP1 CLOSED. Change JP1 only with all power off.

The diode blocks backfeed into the buck; it is not a complete two-source power selector or 12V overvoltage protector.

SUPPLY DISTRIBUTION - branch fuses before long cables

ALITOVE 12V / 6A



Supply negative splits to each station J1-2.

*** F0 is recorded, not approved** for unknown wire sizes.

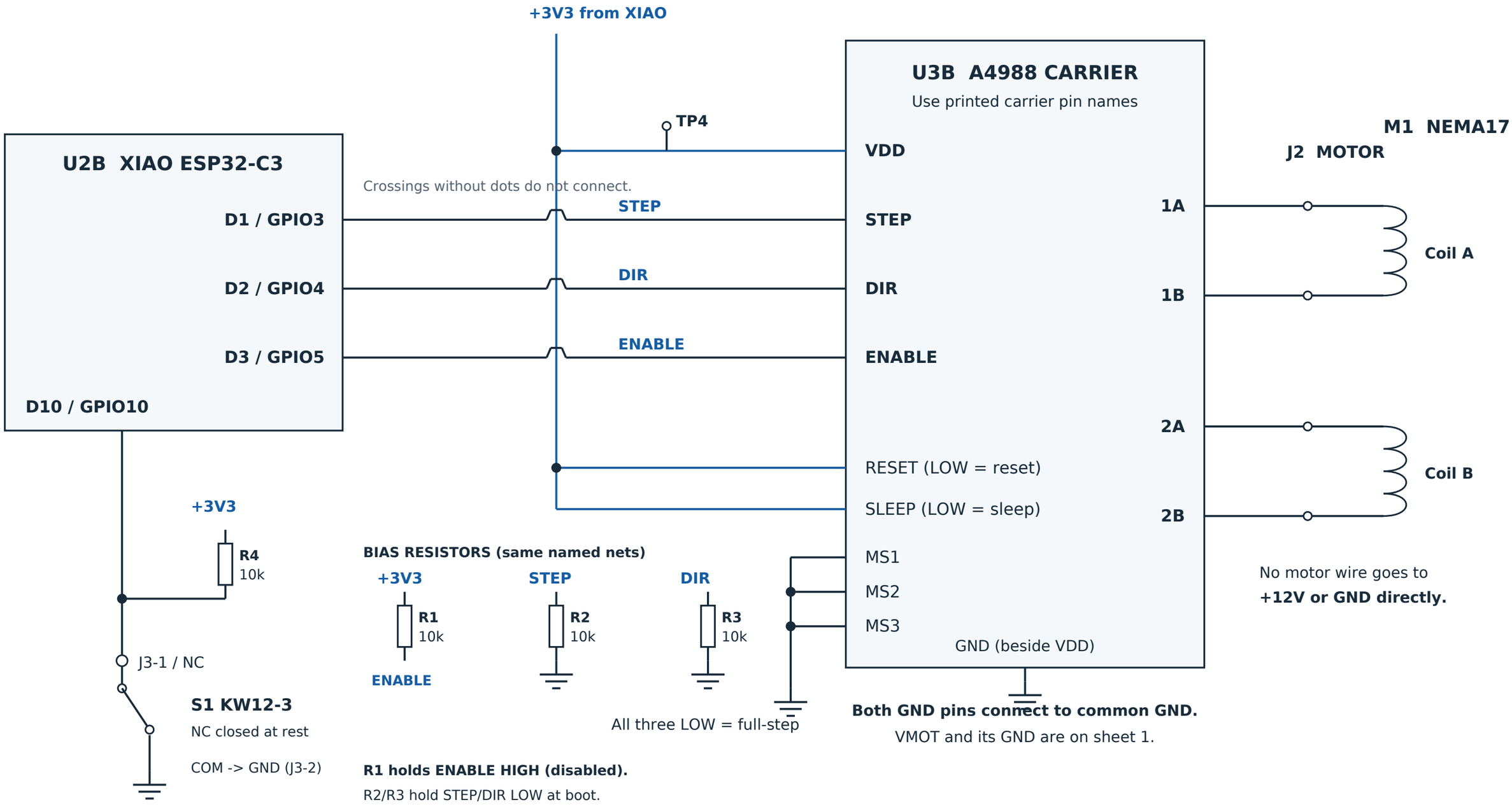
6A total capacity still needs testing.

BUILD DRAWING - check every connection with a meter before first power. Never plug/unplug the motor with power on.

Symbols are functional; use module pin labels, not drawing position.

Control circuit, switch, and motor

U2B and U3B are the SAME physical boards as U2A/U3A on sheet 1. No extra ESP32 or driver is implied.



NO unused. INPUT_PULLUP in firmware.

R4 is an added external pull-up; the earlier test used only the ESP32 internal pull-up. Short/open switch detection is not equivalent:

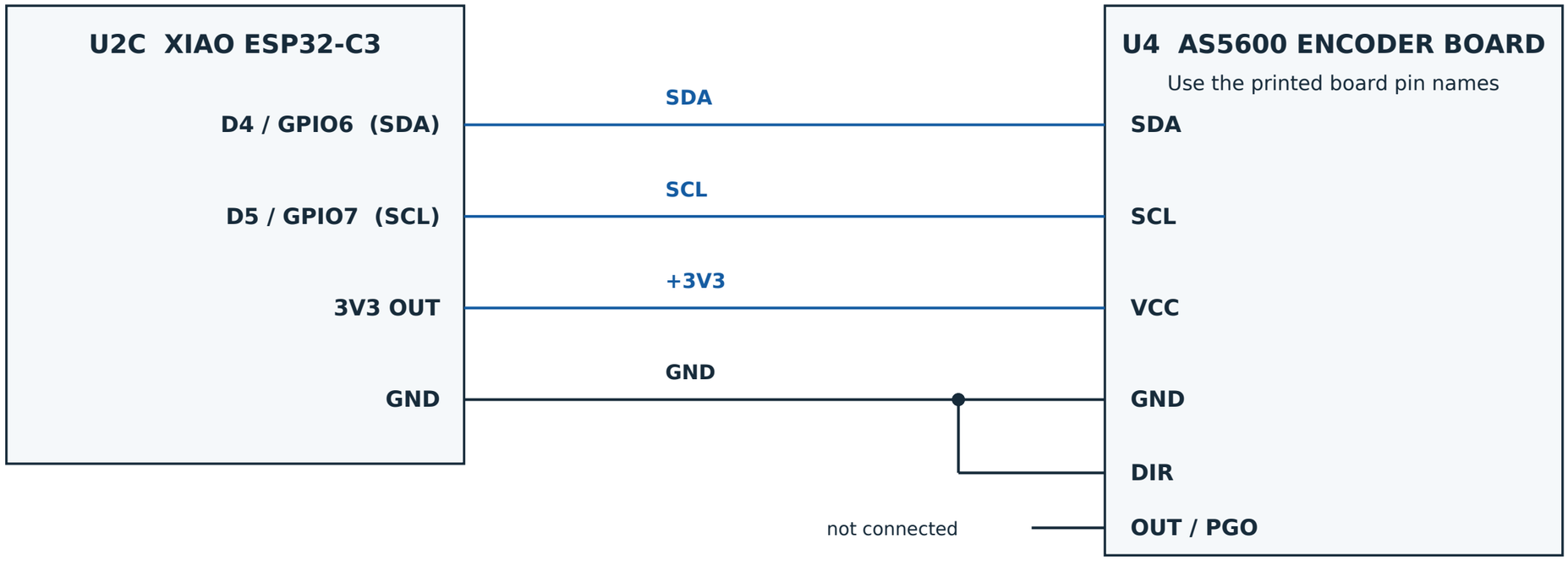
open wire reads STOP; a short to GND can falsely read READY. Test both switch states and unplug detection before motion.

BUILD DRAWING - check every connection with a meter before first power. Never plug/unplug the motor with power on.

Symbols are functional; use module pin labels, not drawing position.

Encoder, motor current (1.35 A) and microstep

Rev D additions. U2C / U3C are the SAME XIAO and A4988 as sheets 1-2. Encoder I2C uses the pins Rev C reserved.



Address 0x36, I2C 400 kHz (firmware).

Pull-ups must go to 3.3 V only.
Board pull-ups (usually 4.7k) to its VCC are fine because VCC = 3.3 V. Never 5 V on ESP32 pins.

If the board has a 3.3V / 5V jumper or **solder pad, set it to 3.3 V.**

DIR to GND fixes the count direction; firmware enc_sign handles the sign.

Magnet: diametric disc on the spool axis, about 0.5-3 mm above the chip, centred.

Keep SDA/SCL/3V3/GND short (< 20 cm), twisted together, away from motor wires.

A4988 CURRENT LIMIT - DECISION D29: 1.35 A (vented shells)

Vref = 8 x I x Rsense I = 1.35 A

SENSE RESISTOR MARKING	Rsense	SET Vref TO
R100	0.100 ohm	1.08 V
R050	0.050 ohm	0.54 V
R200	0.200 ohm	2.16 V

Measure: black probe on driver GND, red on the trim pot (or Vref pad).
12V on, motor DISCONNECTED, logic powered. Turn the pot in small steps.
Read both small resistors near the motor pins before choosing a row.
Heatsink stays on. Confirm with the 60 min hold test:
driver heatsink < ~70 C, else drop to 1.25 A (R100: 1.00 V).

MICROSTEP JUMPERS (MS1 / MS2 / MS3)

MODE	MS1	MS2	MS3	config
Full step (current config)	GND	GND	GND	1
1/16 step (smoother)	+3V3	+3V3	+3V3	16

Hardware and config MUST match: the host converts cable length **to steps with config "microsteps". Mismatch = wrong lengths.**

Full step (all LOW) is what config/robot.example.json uses now.

If you solder MS pins to +3V3, change "microsteps" to 16 for that station before the first run.

BUILD DRAWING - check every connection with a meter before first power. Never plug/unplug the motor with power on.

Symbols are functional; use module pin labels, not drawing position.

Connection ledger and meter checks

This sheet is the exact connectivity reference. Net names repeat across sheets. Measure at labeled test points rather than crowded pins.

NET / FUNCTION	ALL CONNECTED POINTS
+12_STATION	J1-1; buck IN+; A4988 VMOT; C1 positive; TP1
GND / 0V	J1-2; buck IN-/OUT-; XIAO GND; both A4988 GND pins; C1 negative; TP0
GND (continued)	A4988 MS1/MS2/MS3 (full step); R2/R3 lower ends; switch COM (J3-2); AS5600 GND and DIR
SDA / SCL	XIAO D4 / GPIO6 to AS5600 SDA; XIAO D5 / GPIO7 to AS5600 SCL (3.3 V pull-ups only)
+5_BUCK	Buck OUT+; D1 anode (unstriped); TP2
DIODE OUTPUT	D1 cathode (striped); JP1 buck-side contact
XIAO_5V	JP1 XIAO-side contact; XIAO 5V/VBUS; TP3 (USB VBUS is on this board net)
+3V3	XIAO 3V3 OUT; A4988 VDD/RESET/SLEEP; R1/R4 upper ends; AS5600 VCC; TP4
STEP	XIAO D1 / GPIO3; A4988 STEP; R2 upper end
DIR	XIAO D2 / GPIO4; A4988 DIR; R3 upper end
ENABLE	XIAO D3 / GPIO5; A4988 ENABLE; R1 lower end
HOME_NC	XIAO D10 / GPIO10; switch NC (J3-1); R4 lower end
MOTOR OUTPUTS	1A/1B to one measured coil pair; 2A/2B to the other. No common rail connection.

POWER MODES - JP1 disconnects only the external positive feed; ground remains common.

MODE	USB	12V	JP1	RESULT
USB logic / upload	Connected	Off	OPEN	XIAO + driver logic powered; VMOT off
USB motor bench test	Connected	On	OPEN	USB powers logic; 12V powers motor and unloaded buck
Standalone station	Unplugged	On	CLOSED	Buck powers logic; 12V powers motor
Not allowed in this revision	Connected	On	CLOSED	Two sources meet on VBUS; do not use this configuration

DC VOLTAGE CHECKS: black probe at TP0 (J1 negative), red at the selected test point.

TP1: ~12V with supply on. TP2: 5.00V before diode. TP3: ~USB voltage in USB mode, lower than TP2 through diode standalone.

TP4: ~3.3V. RESET/SLEEP: ~3.3V. ENABLE: ~3.3V idle, near 0V during jog. HOME: near 0V released, ~3.3V pressed/open.

A normal meter cannot reliably resolve the short STEP pulse or the 0.4-second ENABLE window. Use scope/logic analyzer if needed.

BUILD DRAWING - check every connection with a meter before first power. Never plug/unplug the motor with power on.

Symbols are functional; use module pin labels, not drawing position.

Audit findings and controlled restart

Schematic reviewed against manufacturer references. Actual perfboard, motor rating, fuse sizing, and suspect components remain unverified.

WHAT IS VERIFIED / WHAT IS NOT

FINDING	DETAIL
Correct topology	12V to VMOT and buck input in parallel; 3.3V to driver VDD/RESET/SLEEP; one common ground.
Reported measurements	VDD, RESET and SLEEP each 3.3V; VMOT-to-adjacent-GND read 0V; later XIAO heated and lost USB recognition.
Diagnosis not established	No proof of short location or failed part. VMOT=0 can be missing feed, missing return, fuse, supply or probe issue.
USB correction	JP1 OPEN for USB. External diode alone does not isolate computer VBUS from all external-power conditions.
Protection limitation	DROK family lists no output overvoltage or input reverse-polarity protection. D1 is not an overvoltage clamp.
New items shown	JP1 disconnect and R2/R3/R4 = 10k resistors are proposed additions; presence in your build is not confirmed.
Vref set by D29	1.35 A: R100 -> 1.08 V, R050 -> 0.54 V (sheet 3). Read both sense resistors; confirm with the hold test.
No approved fuse value	F0=7.5A is existing stock. F1-F4 must protect branch wire/connectors; select with current and fuse-curve data.
Rev D additions	AS5600 on D4/GPIO6 (SDA) + D5/GPIO7 (SCL) at 3.3 V; Vref for 1.35 A (D29); MS jumpers must match config microsteps.

RESTART SEQUENCE - leave the suspect XIAO disconnected throughout passive checks.

1. Remove USB and 12V. Confirm capacitor voltage has fallen near 0V before resistance tests; do not short it with a tool.
2. Isolate/remove suspect XIAO and driver. Check bare wiring against sheet 3; inspect both sides for bridges and shared copper strips.
3. Check intended continuity, including both driver ground socket contacts to J1-2. Check separate power rails for unintended shorts.
4. With electronics disconnected, test supply before/after each fuse against supply negative. Test buck alone and verify 5.00V at TP2.
5. Use a current-limited lab supply for fault finding if available. A 7.5A fuse is not fast semiconductor protection or overvoltage protection.
6. Verify a known-good XIAO separately. JP1 open: add checked logic wiring/driver incrementally, powering off before every change.
7. With EN held HIGH, verify TP1/TP4 and set Vref from confirmed motor/Rsense data. Never guess based on screw direction.
8. Power off; connect checked coil pairs. Run only the short, unloaded jog and switch test. No suspended load or room operation.

SOURCE REFERENCES (manufacturer documentation; accessed 2026-10-16 America/Chicago)

Seeed C3 pin map and board schematic https://wiki.seeedstudio.com/XIAO_ESP32C3_Getting_Started/

A4988 carrier guidance / Allegro IC datasheet https://www.pololu.com/product/1182 | https://www.pololu.com/file/0J450/A4988.pdf

DROK matching module family (exact revision unverified) https://www.droking.com/10pcs-dc-buck-converter-5v-voltage-regulator

BUILD DRAWING - check every connection with a meter before first power. Never plug/unplug the motor with power on.

Symbols are functional; use module pin labels, not drawing position.